PRE-RELEASE

eCH-0309 – Auxiliary document: animal movement data

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Summary

This document describes the data of the animal movement database (TVD) semantically, so that specialists without in-depth IT knowledge can also gain a better understanding of the data.

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Note

This document uses a gender-neutral formulation when referring to persons. This is based on the guidelines (German) of the Federal Chancellery. Depending on the situation, paired forms (citizens), gender-abstract forms (insured person), gender-neutral forms (insured person) or paraphrases without personal reference are used. The generic masculine (citizen) is not permitted. Full forms are used in continuous texts, i.e. in texts consisting of formulated sentences. Short forms can be used in abbreviated text passages, namely in tables. The short form is used with a slash but without an ellipsis (referent). Gender asterisks and similar spellings are not used.

1 Introduction

1.1 Status

In progress: Use is permitted only within the technical group and the Experts' Committee.

1.2 Field of application

This document describes the data of the animal movement database (TVD) semantically, so that specialists without in-depth IT knowledge can also gain a better understanding of the data.

1.3 Animal movement reporting in Switzerland

In the 1990s, Swiss agriculture and the food industry were severely challenged by the animal disease BSE. When the realisation that the disease could be transmitted from animals to humans was confirmed in 1996, this caused great uncertainty. Expectations regarding the safety of food of animal origin rose as a result, as did awareness of the importance of food control. The approach «from stable to table» became established. Risk-based surveillance, seamless traceability of farm animals, control of animal movements, the obligation to declare origin and quality controls throughout the production and processing chain have formed the basis for a high standard of food safety ever since.

The BSE experience led to the establishment of a national animal movement database (TVD) in 1999. Article 7a and Articles 13–15 of the Animal Diseases Act and Title 2, Sections 1, 1a and 2a of the Animal Diseases Ordinance set out the basic legal provisions on the identification/marking and the registration of farm animals as well as the associated reporting obligations. They follow European legislation on the same subject (Regulation (EU) 2016/429 on transmissible animal diseases; Delegated Regulation (EU) 2019/2035 on establishments keeping terrestrial animals and hatcheries, and on traceability).

While only cattle were subject to reporting from 1999, pigs and equids followed from 2011, although pigs are to this day not reported as individual animals. From 2014 and 2020 respectively, the obligation to report to the TVD also applied to small ruminants (sheep and goats). At the beginning, in 2014, this concerned only the slaughterhouses. Since 2014 there has also been an obligation to report poultry. Since January 2020, keepers of small ruminants have been included in individual animal reporting as well. The identification and marking of cloven-hoofed animals with ear tags, the individual registration of cattle, sheep and goats in the TVD, and the reporting of animal arrivals, animal departures (an animal leaves a holding alive), births, imports, exports, slaughter, on-farm slaughter and death (an animal is deregistered as no longer living) are mandatory for all animal keepers in Switzerland.

The TVD is part of a system landscape spanning agriculture, veterinary affairs and food safety. The central entry point is the Agate portal, with the agricultural policy information system AGIS at its centre.

The Ordinance on Identitas AG and the animal movement database (IdTVD-V) further sets out the content relating to animal movements, the tasks of Identitas AG and the rights of access to the data. The importance of identifying/marking farm animals goes far beyond the subject of animal disease and therefore also found its way into agricultural legislation (Agriculture Act LwG, Articles 165g, 177 and 185) and various associated ordinances. The animal movement data have become an indispensable component in implementing agricultural policy measures, for example animal-related direct payments,

the determination of nutrient flows or agri-environmental monitoring. The data also serve as a basis for the structural survey of Swiss agriculture and for agricultural statistics in general.

Animal movement reporting quickly became established along the entire food value chain as well. Programmes with animal welfare, traceability and origin commitments rely on it. The variety of animal information from animal movement reporting is also reflected in the publicly available datasets of the Tierstatistik open data platform.

The technical interface description in this document is based on the technical service description AnimalTracing v1.32 and on the openAPI specification v. x.xx.

1.4 References

Interfaces: AnimalTracing API, technical service description AnimalTracing

NB: the names of the classes described in chapter 3 do not match the ServiceOperations in AnimalTracing, because the semantic description is intended to provide an overview of the data content and is more detailed than the service description.

2 Namespaces

Table 1 lists the prefixes and namespaces used in this document. A prefix stands for a namespace IRI, the common beginning of the IRIs of a vocabulary. The Turtle specification describes how such *prefixed names* are resolved [Turtle].

Table 1: Namespaces used in eCH-0309 – Auxiliary document: animal movement data.

Prefix	Namespace
:	https://agriculture.ld.admin.ch/eCH-0309/1/
xsd:	http://www.w3.org/2001/XMLSchema#

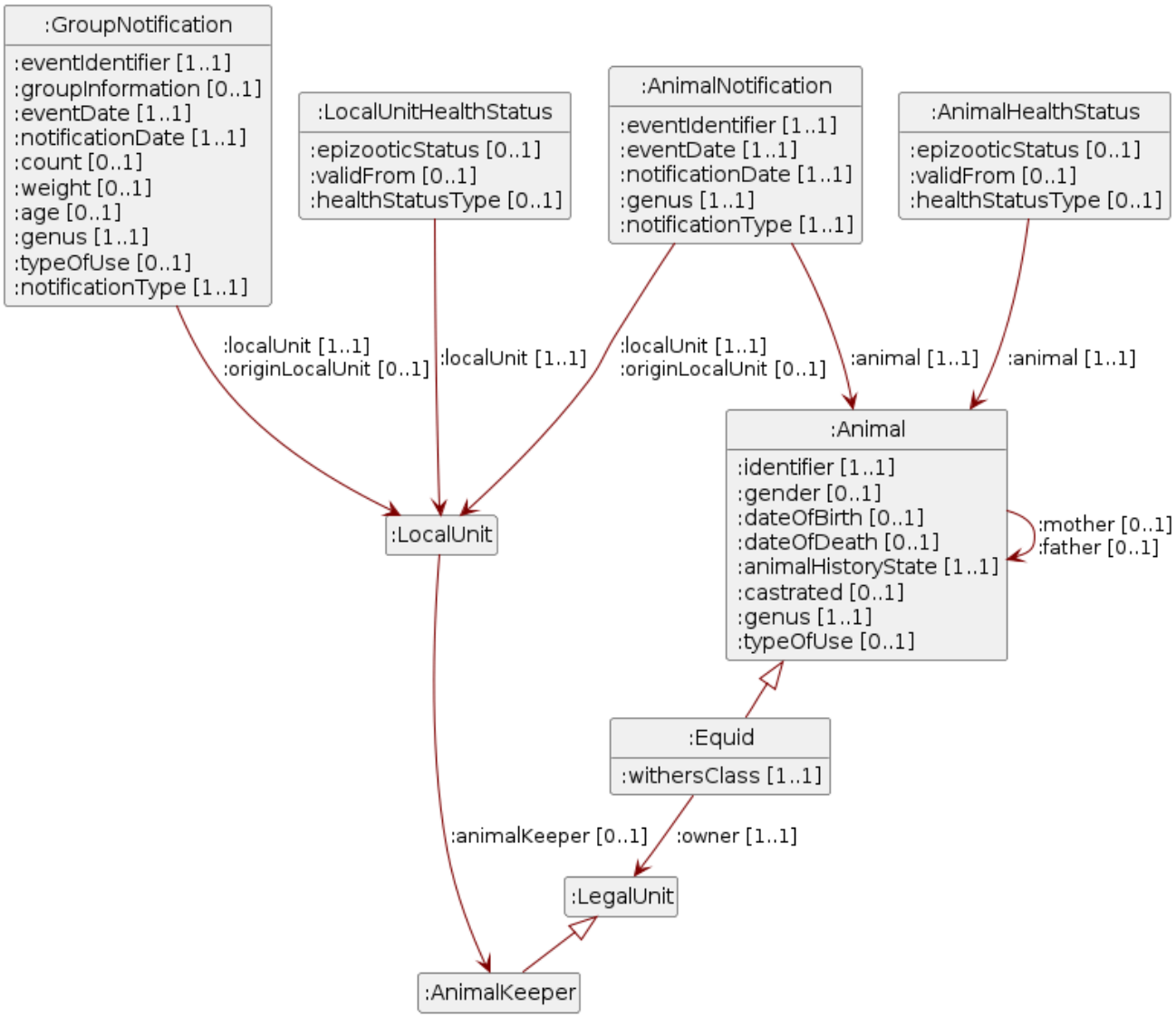


Figure 1: UML diagram of the eCH-0309 data model.

3 Data Model

3.1 Animal keeper

A farm manager (Bewirtschafter) is a legal unit as defined in eCH-0108 that bears the business risk for a farm (LBV Art. 2). The animal keeper is a form of farm manager that keeps animals (LBV Art. 11a) and therefore requires a local unit (animal holding). Responsibility for reporting lies with the animal keeper for all animal categories except equids; for equids it lies with the owner (chapter 3.1.1 of the tool). The attributes are described in eCH-0261, chapters 3.1 and 3.2.

3.1.1 Overview

Shape	:AnimalKeeperShape
Target class	:AnimalKeeper
Closed	No (further properties are allowed)

Data access None (transactional data, not part of the published graph)

3.1.2 Rules (SPARQL)

3.1.2.1 An animal keeper requires at least one local unit (chapter 3.1.1 of the tool). This rule is taken from the semantic description, not from the object catalogue.

```
SELECT $this
WHERE {
    FILTER NOT EXISTS { ?localUnit :animalKeeper $this . }
}
```

3.2 Equid

Subclass of the class individual animal for animals of the equid genus (horse, mule, hinny, donkey). All attributes of the class individual animal also apply to equids; this class describes only the additional, equid-specific information. For equids, responsibility for reporting lies with the owner rather than with the animal keeper. An owner is the legal unit that owns equids; the term is used for equid owners only (chapter 3.1.1 of the tool).

3.2.1 Overview

Shape :EquidShape
Target class :Equid
Closed No (further properties are allowed)
Data access None (transactional data, not part of the published graph)

3.2.2 Properties

Description	Details	Cardinality
Owner (:owner): The legal unit that owns the equid. For equids, responsibility for reporting lies with the owner rather than with the animal keeper (chapter 3.1.1 of the tool).	• Type: Legal unit	1..1
Withers class (:withersClass)	• Type: :EquidWithersClass	1..1

3.3 Group notification

An event concerning a group of animals that cannot be identified individually. Group notifications are recorded for animals of the categories pigs and poultry; the individuals of such a group are therefore not known. Events are reported by the natural persons of the responsible legal unit, the animal keeper. They comprise housing notifications for poultry, movement notifications for pigs, and slaughter notifications for pigs (by head) and poultry groups (in kg). This class has no counterpart in the AnimalTracing openAPI specification, because pigs and poultry are not covered there.

3.3.1 Overview

Shape	:GroupNotificationShape
Target class	:GroupNotification
Closed	No (further properties are allowed)
Data access	None (transactional data, not part of the published graph)

3.3.2 Properties

Description	Details	Cardinality
Local unit (:localUnit): BUR number of the local unit	• Type: Local unit	1..1
Origin (:originLocalUnit): BUR number of the local unit the animal comes from	• Type: Local unit	0..1
ID (:eventIdentifier): Unique identifier of the movement, assigned by the system.	• Type: Text (xsd:string)	1..1
Genus (:genus)	• Type: :Genus	1..1
Group information (:groupInformation): Group information must be given for housing notifications of poultry	• Type: Text (xsd:string)	0..1
Notification type (:notificationType)	• Type: :NotificationType	1..1
Event date (:eventDate): Date on which the event took place	• Type: Date (xsd:date)	1..1
Notification date (:notificationDate): Date on which the event was reported	• Type: Date (xsd:date)	1..1
Number (:count): Number of animals moved in one movement (pigs/poultry)	• Type: Integer (xsd:integer)	0..1
Weight (:weight): Total weight in kg of the animal group for slaughter notifications of poultry	• Type: Integer (xsd:integer)	0..1
Type of use (:typeOfUse): Type of use of the animal group (for individual animals it is held on the class «Einzeltier»)	• Type: :AnimalTypeOfUse	0..1
Age (:age): For housing notifications of poultry the age must be given in weeks	• Type: Integer (xsd:integer)	0..1

3.4 Health status of a local unit

Where animal disease events are of national significance, it is essential that animal keepers are informed about movement bans and disease status, so that the risk of spread through animal movements can be minimised. Disease information recorded for a local unit. In addition to the epizootic status of a local unit, vaccination, risk and laboratory results can also be represented. The AnimalTracing openAPI specification has no dedicated class for this; there the corresponding information is held as attributes of the local unit.

3.4.1 Overview

Shape	:LocalUnitHealthStatusShape
Target class	:LocalUnitHealthStatus
Closed	No (further properties are allowed)
Data access	None (transactional data, not part of the published graph)

3.4.2 Properties

Description	Details	Cardinality
Local unit (:localUnit): BUR number of the local unit	• Type: Local unit	1..1
Type (:healthStatusType): e.g. epizootic, vaccination or risk status	• Type: :HealthStatusType	0..1
Status (:epizooticStatus)	• Type: :LocalUnitEpizooticStatus	0..1
Date (:validFrom): Date from which the status is valid	• Type: Date (xsd:date)	0..1

3.5 Health status of an individual animal

Disease information recorded for an individual animal. In addition to the individual epizootic status, relevant vaccinations are recorded as well.

3.5.1 Overview

Shape	:AnimalHealthStatusShape
Target class	:AnimalHealthStatus
Closed	No (further properties are allowed)
Data access	None (transactional data, not part of the published graph)

3.5.2 Properties

Description	Details	Cardinality
Animal ID (:animal): A health status of an individual animal refers to exactly one individual animal.	• Type: Individual animal	1..1
Type (:healthStatusType): e.g. epizootic, vaccination or risk status	• Type: :HealthStatusType	0..1
Status (:epizooticStatus)	• Type: :AnimalEpizooticStatus	0..1
Date (:validFrom): Date from which the status is valid, or the date of vaccination	• Type: Date (xsd:date)	0..1

3.6 Individual animal

An individually identified farm animal. In the animal movement database, animals of the categories cattle, sheep, goats and equids are recorded as individual animals; they are uniquely identified by their ear tag

number, or by the UELN in the case of equids. No individual animals are kept for pigs and poultry, only group notifications.

3.6.1 Overview

Shape	:AnimalShape
Target class	:Animal
Closed	No (further properties are allowed)
Data access	None (transactional data, not part of the published graph)

3.6.2 Properties

Description	Details	Cardinality
Mother (:mother): An individual animal has at most one mother animal.	• Type: Individual animal	0..1
Father (:father): An individual animal has at most one father animal.	• Type: Individual animal	0..1
Identifier (:identifier): Ear tag number, or UELN for equids	• Type: Text (xsd:string) • Length: ≤ 15 characters	1..1
Genus (:genus)	• Type: :Genus	1..1
Gender (:gender)	• Type: :Gender	0..1
Date of birth (:dateOfBirth)	• Type: Date (xsd:date)	0..1
Date of death (:dateOfDeath)	• Type: Date (xsd:date)	0..1
Animal history state (:animalHistoryState)	• Type: :AnimalHistoryState	1..1
Type of use (:typeOfUse)	• Type: :AnimalTypeOfUse	0..1
Castrated (:castrated)	• Type: Boolean (xsd:boolean)	0..1

3.7 Individual animal notification

An event concerning an individual animal. Events comprise movement notifications, notifications of basic data, changes to individual attributes of the basic data as well as final notifications. A notification always refers to the local unit at which the event took place and may in addition state the local unit of origin. Events are reported by the authorised persons, stating the animal holding to which the notification refers; for equids they are reported by the authorised persons of the responsible equid owner.

3.7.1 Overview

Shape	:AnimalNotificationShape
Target class	:AnimalNotification
Closed	No (further properties are allowed)
Data access	None (transactional data, not part of the published graph)

3.7.2 Properties

Description	Details	Cardinality
Identifier (:animal): An individual animal notification refers to exactly one individual animal.	• Type: Individual animal	1..1
Local unit (:localUnit): BUR number of the local unit to which the notification refers	• Type: Local unit	1..1
Origin (:originLocalUnit): BUR number of the local unit the animal comes from	• Type: Local unit	0..1
ID (:eventIdentifier): Unique identifier of the movement, assigned by the system.	• Type: Text (xsd:string)	1..1
Genus (:genus)	• Type: :Genus	1..1
Notification type (:notificationType)	• Type: :NotificationType	1..1
Event date (:eventDate): Date on which the event took place	• Type: Date (xsd:date)	1..1
Notification date (:notificationDate): Date on which the event was reported	• Type: Date (xsd:date)	1..1

3.8 Legal unit

A legal unit as defined in eCH-0108 is a legal entity (e.g. a public limited company or a limited liability company), a partnership (e.g. a general partnership) or a self-employed natural person that is subject to the federal act on the business identification number (UIDG) and has an entry in the UID register. It identifies, for example, the taxable entity for the tax authorities or the person liable for social insurance contributions. The legal unit is identified by its UID. Only where the enterprise is a «sole proprietorship» is the farm manager at the same time a natural person. The master system for this data is the FSO's UID register; the agricultural information system obtains the data from that register through an interface. The animal keeper and the equid owner are the two forms of legal unit that occur in this Hilfsmittel (chapter 3.1.1 of the tool). The attributes are described in eCH-0261, chapters 3.1 and 3.2, and in eCH-0108, chapter 3.1.

3.8.1 Overview

Shape	:LegalUnitShape
Target class	:LegalUnit
Closed	No (further properties are allowed)
Data access	None (transactional data, not part of the published graph)

3.9 Local unit

A local unit as defined in eCH-0108 is a facility at a given location where an economic activity is carried out. A local unit has a BUR number and is always assigned to a legal unit (or enterprise). The local unit is identified by its BUR number. A local unit may consist of one or more buildings. The EGID assigned to the BUR number refers to «the centre» of the local unit, that is, its main building. In the TVD, local units are called «animal holdings» and are additionally identified by their TVD number. This definition is taken

from eCH-0108, chapter 1.5; the agriculture-specific additions are described in eCH-0261, chapters 3.3 and 3.4.

3.9.1 Overview

Shape	:LocalUnitShape
Target class	:LocalUnit
Closed	No (further properties are allowed)
Data access	None (transactional data, not part of the published graph)

3.9.2 Properties

Description	Details	Cardinality
Animal keeper (:animalKeeper): Assignment of the local unit to the animal keeper. According to eCH-0108, chapter 1.5, a local unit is always assigned to a legal unit (or enterprise); the master data of the local unit carries the optional attribute «mainUid» holding the UID of the main legal unit. Because the object catalogue treats the animal keeper only as a special form of the legal unit, this assignment is not mandatory.	• Type: Animal keeper	0..1

4 Code lists

For some of these code lists no French or Italian designations are available yet; the designations below are given in German and English, as provided by the sources. (https://test-03-tvd-api-at.identitas.ch/open-api/v1.0, 2026-09-24)

4.1 Health status of an individual animal

Table 2: Values of the code list Health status of an individual animal

Value	Designation	Description
`Blocked`	Blocked	Type: epizootic status
`Free`	Free	Type: epizootic status
`NotTested`	Not tested	Type: epizootic status
`Vaccinated`	Vaccinated	Type: vaccination status

4.2 Animal history state

Table 3: Values of the code list Animal history state

Value	Designation	Description
`Lost`	Lost	
`NotDefined`	Not defined	
`NotOk`	Not ok	
`Ok`	Ok	
`TemporaryOk`	Temporary ok	

4.3 Type of use (cattle, sheep, goats)

Table 4: Values of the code list Type of use (cattle, sheep, goats)

Value	Designation	Description
`Milk`	Milk	Milchkühe, -schafe und -ziegen
`Other`	Other	

4.4 Type of use (equids)

Table 5: Values of the code list Type of use (equids)

Value	Designation	Description
`CompanionAnimal`	Companion animal	
`FarmAnimal`	Farm animal	

4.5 Withers class

Table 6: Values of the code list Withers class

Value	Designation	Description
`GreaterThan148cm`	Withers height above 148 cm	
`LessOrEqualThan148cm`	Withers height up to 148 cm	

4.6 Gender

Table 7: Values of the code list Gender

Value	Designation	Description
`Female`	Female	
`Male`	Male	

4.7 Genus

Table 8: Values of the code list Genus

Value	Designation	Description
`Camelid`	Camelid	Neuweltkameliden
`Cattle`	Cattle	Tiere der Rindergattung (Bos) und Wasserbüffel (Bubalus bubalis)
`Equid`	Equid	Tiere der Pferdegattung (Pferd, Maultier, Maulesel, Esel)
`Game`	Game	Wild in Gehegen
`Goat`	Goat	
`Pig`	Pig	
`Poultry`	Poultry	
`Sheep`	Sheep	

4.8 Health status type

Table 9: Values of the code list Health status type

Value	Designation	Description
`EpizooticStatus`	Epizootic status	
`LaboratoryResult`	Laboratory result	
`RiskStatus`	Risk status	
`VaccinationStatus`	Vaccination status	

4.9 Health status of a local unit

Table 10: Values of the code list Health status of a local unit

Value	Designation	Description
`Blocked`	Blocked	Type: epizootic status
`Free`	Free	Type: epizootic status
`High`	High	Type: risk status
`Low`	Low	Type: risk status
`Medium`	Medium	Type: risk status
`Negative`	Negative	Type: laboratory result
`NotTested`	Not tested	Type: epizootic status
`Positive`	Positive	Type: laboratory result

4.10 Notification type

Table 11: Values of the code list Notification type

Value	Designation	Description
`Arrival`	Arrival	Not for equids
`Birth`	Birth	
`DayStay`	Day stay	Not for equids
`DeathBirth`	Stillbirth	Not for equids
`Deceased`	Death	For equids: euthanasia
`Export`	Export	For equids: transfer of ownership abroad
`FirstRegistration`	First registration	
`Import`	Import	
`ImportAfterExport`	Import after export	Not for equids
`Leaving`	Departure	Not for equids
`LocationChange`	Change of location	For equids only
`OnFarmSlaughter`	On-farm slaughter	Not for equids
`Slaughter`	Slaughter	

5 Safety considerations

Information about the explicitly relevant legal bases or a note that during the implementation the relevant legal bases must be observed.

6 Disclaimer

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8 Annex A - References

[Turtle] Beckett, David ; Berners-Lee, Tim ; Prud'hommeaux, Eric ; Carothers, Gavin: *RDF 1.1 Turtle – Terse RDF Triple Language* (W3C Recommendation) : World Wide Web Consortium (W3C), 2014

9 Annex B - Cooperation and Verification

10 Annex C - Abbreviations and Glossary

Table 12: Glossary of eCH-0309

Term	Description
Animal holding	In the animal movement database, local units are called «animal holdings»; they are additionally identified by their TVD number.
Farm manager	A legal unit that bears the business risk for a farm (LBV Art. 2). The animal keeper is a form of farm manager that keeps animals (LBV Art. 11a).
Home holding	For cattle, a home holding can be recorded in order to represent an animal's affiliation to an animal holding while the animal is located on another local unit.
Linked Data Service (LINDAS)	The official Linked Data service of the Swiss Federal Administration, acting as a triple store.
Reporting person	The person or persons who make notifications for an animal holding or for an equid owner.
Resource Description Framework (RDF)	A central standard of the World Wide Web Consortium (W3C) for modeling data structures on the Web. Information is represented as networked graphs rather than in traditional tables.
Triple	The basic structure of a statement in RDF, consisting of a subject, predicate, and object. *Broader*: Resource Description Framework

11 Annex D - Changes in comparison to previous version

12 Annex E - Table of figures

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